Venkata Lakshmi Parimala Pasupuleti

AI Solutions Architect · Explainable AI · Confidence Scoring · Runtime Governance · RAG Pipelines · LLMOps · Python

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PROFESSIONAL SUMMARY

AI systems builder with 8+ years of production engineering at Entrata, specializing in **explainable AI, confidence scoring, and runtime governance** — the three core primitives of Meibel's platform. Shipped Entrata's first production AI pipeline across 1.6M financial records; delivered 20+ modules for 100K+ monthly active users across federal compliance programs (LIHTC, HUD, HAP, AMI); translated complex business requirements for enterprise customers into scalable production architecture across Compliance, Legal, Operations, and Product. Built **Leasel**: a runtime AI governance platform (confidence-gated HITL, citation-enforced explainability, deterministic compliance gates) — every AI decision traceable, every regulated call human-reviewed, every answer grounded in source or explicitly flagged. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Explainable AI | Confidence Scoring | Runtime Governance | RAG Workflow Design | LLMOps & Observability | Model Evaluation | Customer-Facing AI Deployment | PoC to Production Rollout | Data Governance & Compliance | Cross-Functional Technical Delivery | Python & AWS | Prompt Engineering

EXPERIENCE

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

Research · Statistical Modeling · Python Instruction

- Built prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection surfacing issues manual review missed; mentored 600+ students in Python programming and database design.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Production AI · Data Governance · Cross-Functional Delivery at Scale

- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) — automated complex workflows with output validation, quality gates, and data governance controls enforced from day one; zero violations at launch.
- **Cross-functional delivery across 120+ Agile sprints** for **enterprise customers** — translated federal housing law (LIHTC, HUD, HAP, AMI, Section 8) into production architecture; led **customer onboarding and integration** across new compliance modules; created **architectural diagrams and documentation** (Lucidchart) for schema redesigns and service boundaries; acted as the technical bridge between Compliance, Legal, Operations, and Product.
- **Data governance and compliance at scale**: 100% PHI compliance on AWS S3 migration (Libsodium-encrypted; zero security incidents); zero compliance violations across all federal housing programs; designed deployment strategies for microservices architecture redesign (Docker, 50% scalability improvement).
- **Troubleshot and resolved** complex system issues across data, API, and cloud layers — built internal debugging and performance tools reducing debugging time 40%; maintained bug count below 10 for 6+ consecutive months; optimized 200+ SQL queries (30% gain) through execution plan analysis and indexing strategy.
- Eliminated 80% of manual operational workload via Python/Bash automation; optimized 200+ SQL queries (30% performance gain); mentored 5+ engineers across 300+ peer code reviews.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Backend · Data Pipelines · AWS S3

- Built backend compliance modules (PHP, React, PostgreSQL, JavaScript); processed 1.6M HUD/Tax Credit records; migrated client documents to AWS S3; redesigned legacy database architecture.

PORTFOLIO PROJECTS

Leasel — Runtime AI Governance Platform: Confidence Scoring, Explainability & HITL

Explainable AI | Runtime Governance | Confidence Scoring

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built on the same three primitives as Meibel's platform — **confidence scoring, runtime governance, explainability**: an 8-agent LangGraph StateGraph where every AI decision is routed based on confidence threshold, every answer is cited to an exact source clause or flagged and downgraded, and every regulated eligibility call is made by deterministic logic or a human — never the AI alone.
- **Explainability enforced in architecture**: 100% citation correctness, 0% unsupported-claim rate; customers see exactly which source clause drove each decision; 0 safety boundary violations on full validation set; tested for adversarial inputs in tenant-facing surface.
- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs (5 US states + federal); LLM-as-judge scoring with multi-trial sampling and regression gate — production-grade eval baked in, not bolted on.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

AI Architecture & Governance: Explainable AI, Confidence Scoring, Runtime Governance, HITL, Human-in-the-Loop, Guardrails, Deterministic Compliance Gates, Data Governance, Ethical AI, Production AI, Agentic AI, LLM Orchestration, Multi-Agent Orchestration, LangGraph, LangChain

LLMOps & Evaluation: LLMOps, LLM Evaluation, LLM-as-Judge, Eval Frameworks, CI-Integrated Evals, Agent Observability, Agent Tracing, Drift Detection, RAG Pipelines, Hybrid Retrieval, Semantic Search, Embeddings, ChromaDB, Vector Database, Prompt Engineering, Multi-Provider LLM Routing, Anthropic API, OpenAI API

Backend & Infrastructure: Python (primary), FastAPI, REST APIs, React, TypeScript, JavaScript, AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), PostgreSQL, Microservices, Data Pipelines, ETL, RabbitMQ, Git, Linux, Agile/Scrum

Compliance & Security: PHI Compliance, PII Handling (Libsodium), Federal Regulatory Compliance (LIHTC, HUD, HAP, AMI), Regulated Environment AI, Data Security, Encryption